

(0-2)

Data Analysis (What happens)

* Process of collecting, cleaning, organizing and examining data to find useful information & support decision-making.

Data Analytics (What is likely to happen).

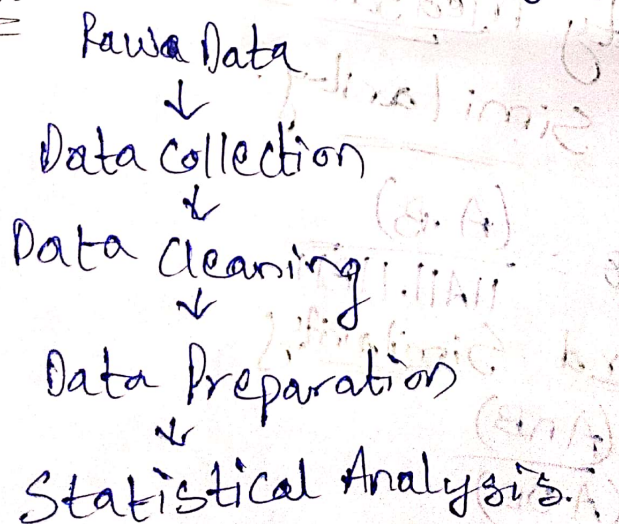
* It is the use of tools & techniques to analyze data & generate insights automatically.

Exploratory Data Science (EDA)

* Process of examining, summarizing, & visualizing datasets to understand their underlying structure & patterns.

- Provides Deeper understanding of Data.
- Identifies data quality issues.
- Detects outliers & anomalies.
- Improve data preprocessing.
- Enhances model accuracy & reliability.

Workflow



↓
Visualization

↓
Pattern Discovery

↓
Feature Engineering.

↓
Insights

Python program for missing value analysis

```
import pandas as pd
```

```
import numpy as np
```

```
data = {
```

```
    'Age': [25, 30, np.nan, 28, 35],
```

```
    'Salary': [50000, np.nan, 60000, 55000, 52000]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
print(df.isnull().sum())
```

⇒ Checks missing values.

```
df['Age'].fillna(df['Age'].mean(),
```

```
                inplace=True)
```

```
df['Salary'].fillna(df['Salary'].median(),
```

```
                    inplace=True)
```

```
print(df)
```

Data Visualization Techniques

- Bar chart = Compare categories.
- Line chart = Show trends over time
- Histogram = Display data distribution
- Scatter Plot = Show relationships b/w variables.
- Box Plot = Detect outliers
- Pie chart = Show proportions.
- Heat Map = Display correlations.

Data Inspection

- * Examine the dataset's dimensions, column names, data types & missing values.

Functions

- `df.head()` ⇒ Displays the first 5 rows
- `df.tail()` ⇒ Displays the last 5 rows
- `df.shape` ⇒ Returns no of rows & columns.
- `df.columns` ⇒ Shows column names.
- `df.dtypes` ⇒ Displays data types of attributes.
- `df.info()` ⇒ Provides dataset information & null values.
- `df.isnull().sum()` ⇒ Counts missing values in each column.

Data Understanding

* Exploring the dataset to identify patterns, relationships, missing values & distributions.

Functions

`df.corr()` $\Rightarrow$ Measures correlation b/w numerical attributes.

`df.duplicated - sum` $\Rightarrow$ Counts duplicate records.

`df.value - counts()` $\Rightarrow$ Counts frequency of unique values.

`df.hist()` $\Rightarrow$ Visualizes feature distributions.

`df.plot()` $\Rightarrow$ Creates graphical representations of data.

Summary Statistics

* Provide numerical descriptions of datasets.

Functions

`df.describe()` $\Rightarrow$ Generates statistical summary.

`df.mean()`

`df.median()`

`df.mode()`

`df.std()` $\Rightarrow$ Standard Deviation

`df.min()`

`df.max()`

`df.var()` $\Rightarrow$ Variance.

Data Visualization

* Graphical representation of data.

Matplotlib

* Used for creating static & interactive plots.

- High-quality visualizations.
- Integrates with NumPy & Pandas.
- Cross-platform support.

import matplotlib.pyplot as plt
↓
gives plotting functions.

Plotting

* Generating visual representations of data to enhance comprehension, analysis, & communication of information.

- Data Exploration.
- Data Analysis
- Communication
- Storytelling.

Seaborn

* Provide of high-level interface for creating informative & attractive statistical graphics.

Plotly

* Versatile library for creating interactive, web-based visualizations.

other libraries ⇒

Ex:- import m

X = [1, 2, 3, 4]

Y = [10, 20, 30, 40]

plt.title

plt.xlabel

plt.ylabel

plt.plot

plt.show

Scatter plot

* Displays the two variable

Seaborn

* This is par
Complex da
more intuit

→ Beautiful

→ Supports

Seaborn

→ High-level. A

→ Attractive th

→ Statistical pl

→ Easy syntax

other libraries $\Rightarrow$ Altair, Holoviews, ggplot,
Geopandas, Mayavi.

Ex: `import matplotlib.pyplot as plt`

`x = [1, 2, 3, 4]`

`y = [10, 20, 25, 30]`

`plt.title("Sales")`

`plt.xlabel("x")`

`plt.ylabel("y")`

`plt.plot(x, y)`

`plt.show()`

Scatter plot

* Displays the relationship between the two variables or datasets.

Seaborn

* This is particularly for visualizing complex datasets & making data analysis more intuitive.

$\rightarrow$ Beautiful built-in themes.

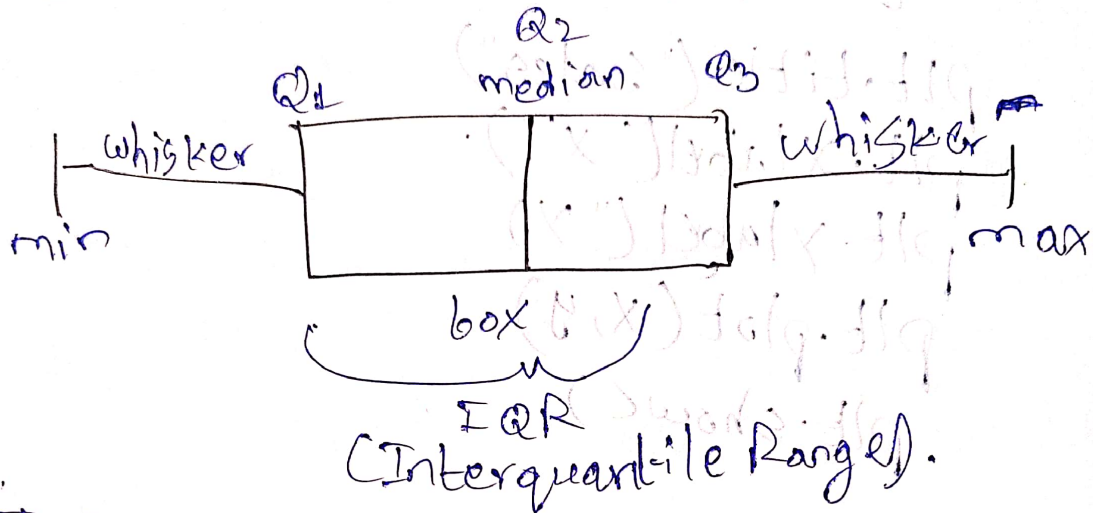
$\rightarrow$ Supports multi-variable visualization.

Seaborn	Matplotlib
$\rightarrow$ High-level API	$\rightarrow$ Low-level API
$\rightarrow$ Attractive themes	$\rightarrow$ Manual styling.
$\rightarrow$ Statistical plots	$\rightarrow$ Basic plotting.
$\rightarrow$ Easy syntax	$\rightarrow$ More coding.

Built-in datasets in Seaborn

- iris
- tips
- titanic
- flights.

Box Plot



$$IQR = Q3 - Q1$$

$Q1 = 25^{\text{th}}$ Percentile

$Q2 = \text{Median}$

$Q3 = 75^{\text{th}}$ Percentile.

$$\begin{aligned} * \text{Lower Bound} &= Q1 - 1.5(IQR) \\ * \text{Upper Bound} &= Q3 + 1.5(IQR) \end{aligned}$$

Methods for detecting outliers

- IQR Method
- Z-score method
- Modified Z-score
- Box Plot Visualization
- DBSCAN Clustering

Z-Score

$$Z = \frac{X - \mu}{\sigma}$$

$Z > 3$ or $Z < -3$ is considered as outlier.

Ex:- 10, 12, 14, 15, 18, 20, 22, 24, 100

Lower half = 10, 12, 14, 15

$Q_2 = 18$

Upper half = 20, 22, 24, 100

$$IQR = 23 - 13 = 10$$

$$Q_1 = \frac{12 + 14}{2} = \frac{26}{2} = 13$$

$$Q_3 = \frac{22 + 24}{2} = \frac{46}{2} = 23$$

$$\begin{aligned} L.B &= Q_1 - 1.5(IQR) & U.B &= Q_3 + 1.5(IQR) \\ &= 13 - 1.5(10) & &= 23 + 1.5(10) \\ &= -2 & &= 38 \end{aligned}$$

$100 > 38 \Rightarrow \text{outlier} = 100$

Finding Quartiles in Python

$$Q_1 = df['A'].quantile(0.25)$$

$$Q_3 = df['A'].quantile(0.75)$$

Correlation

* It is a statistical metric that measures how strongly two variables are related & how they change together.

* -1 to +1. $(-1 \leq r \leq 1)$

Types

- Positive Correlation (Strong/Weak)
- Negative Correlation (Strong/Weak/Moderate)
- Zero/No correlation.

$+1$ = Perfect Positive
 0.7 to 0.9 = Strong Positive
 0.4 to 0.6 = Moderate
 0 = No Correlation
 -0.4 to -0.6 = Moderate Negative
 -0.7 to -0.9 = Strong negative
 -1 = Perfect negative.

Pearson Correlation

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Ex:- $X = [1, 2, 3]$

$Y = [2, 3, 7]$

$\bar{x} = \frac{6}{3} = 2$

$\bar{y} = \frac{12}{3} = 4$

~~$\sum (x_i - \bar{x}) = 0$~~
 ~~$\sum (y_i - \bar{y}) = 0$~~

$r = \frac{5}{\sqrt{14}} = \frac{5}{\sqrt{28}} = \frac{5}{5.2915} = 0.9449$

Spearman Rank correlation coefficient

$$r = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

$d =$ ~~rank~~ Square of the rank difference.

Kendall Correlation

$$\gamma = \frac{\text{Concordant Pairs} - \text{Discordant Pairs}}{\text{Total Pairs}}$$

Concordant = matching.

Discordant = mismatching.

HeatMap

- * Uses colors to represent values.
- * Used to visualize a Correlation Matrix.
- * Used in machine learning.

→ Dark colors = Strong correlation.

→ Light colors = Weak correlation.

Applications

- Sales Analysis
- Climate Studies
- Customer Analytics.

Dashboards

* This displays various types of visual data in one place.

→ Organizations use Dashboards to,

- * Track KPIs
- * Monitor business performance.
- * Identify trends.
- * Improve productivity.
- * Make data-driven decisions.

Tools

BI Tools

→ Power BI

→ Tableau

Web-based

→ dc.js

→ D3.js

→ Chart.js

Cloud-native

→ Google Data

Studio

→ AWS QuickSight

Types of Dashboards

→ Operational (What is happening now)

* Real-time monitoring

→ Tactical (Why is it happening)

* Department-level analysis

→ Strategic (What should we do next)

* Executive-level decision making

Dashboard Architecture

→ Data Sources

→ Data processing

→ Database

→ Visualization Layer

→ User Interface

Workflow

Data Collection



Data Cleaning



Analysis



Visualization



Decision Making